Check in please

Network Security Lab

WiCyS UIC x LUG

START MENU

Agenda

Topics Covered

1) Warnings

2)what is NMap

3) What is Wireshark

4) Hands-on Lab!!!!!!

Warning

- Unauthorized NMap scanning can be seen as a violation of law
- Do **NOT** scan UIC's networks without written permission from IT

UIC IT if you try a NMap scan on their network

Warning

- Won’t learn NMap or Wireshark in depth today
- You would need alot more knowledge about TCP/IP to grasp how this truly works

What is NMap?

(Aka Network Mapper)

Open source security tool for network exploration and security auditing.

YES

NO

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Who uses it?

- Pen testers
- Network admin
- Security analysts
- malicious threat actors
- you guys :0
- etc

YES

NO

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What does it do?

Uses raw IP packets to discover:

- what host are available on network
- what services is the host offering
- what OS system is running (windows,linux, etc.)
- type of packet filters/firewalls are in use
- etc

YES NO

There are MANY different types of scans that we can't cover today!!!!!!!!

What does it do?

Nmap scripting engine (NSE):

- one of the most powerful and flexible features
- can automate a wide veriety of networking tasks
- vulnerability detection and exploitation
- Backdoor detection

YES NO

There are MANY different types of scans that we can't cover today!!!!!!!!

Useful Flags

-sS → fast, stealthy, terminates after receiving TCP Syn-Ack

-sU/-sT → allows for you to scan ports by protocol type(UDP/TCP)

-p → allows you to scan for specific port

YES NO

There are MANY different types of scans that we can't cover today!!!!!!!!

TCP

Transmission Control Protocol(TCP)
Transport layer protocol
reliable connections between devices, ensure delivery in correct order without errors

UDP

User Datagram Protocol(UDP) port
Transport layer protocol
Sends data without establishing a connection, quick, ideal for real-time applications like streaming or gaming

FTP

File Transfer Protocol(FTP)
port #21
Application layer protocol - Uses **TCP**
For uploading and downloading files between a computer and a remote server

There are MANY other protocols like HTTP, SMTP, SCP!

What is Wireshark?

(Aka Packet Sniffer)

Open source network protocol analyzer used for packet capture and analysis

YES

NO

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Who uses it?

- Network admin
- Network security engineers
- QA engineers
- Developers
- you guys :0
- etc

YES

NO

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What does it do?

- Capture live packet data from a network interface and save data to a file
- Analyse pcap/pcapng files
- Display packets with very detailed protocol information
- Export some or all packets in a number of capture file formats
- Filter packets on many criteria
- Search for packets on many criteria
- Colorize packet display based on filters
- Create various statistics

and many more!!

YES NO

There are MANY more features that we can't cover today!!!!!!!!

Useful Display Filters

- ip.addr == <IP_address>
- tcp.port == <port_number>
- ip.dst == <IP_address> (Destination IP)
- ip.src == <IP_address> (Source IP)
- tcp
- ftp
- http
- etc

YES NO

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LAB OVERVIEW

Defender Goal:
Detect the malicious attacker using wireshark & Defend your server by updating the FTP config file so the attacker can’t get in anymore

Harder role!!

LAB OVERVIEW

Attacker Goal:

Find a way to login to the server without knowing the credentials & send the Defender a malicious file “evil.sh”

Time to start lab!!

1. Pair up into groups of two!
2. Use number slip to log in to vm
3. Read through the handout
4. If the attacker wins still try to finish the defender part after!

Proxmox Login

User: user[add number here]

Password: wicyspassword

Set Realm to Proxmox VE

You and your partners login is the same

Thank you!

Please try to get familiar with NMap and Wireshark on your own!!!

YES NO

Want to learn more?

click the links!

NMap official website

Hack the Box

WireShark

TCP/IP

Check in please